Interpretable Transformer Models for Financial Time Series Forecasting: Discovering Emergent Market Dynamics

MSc Data Science and Artificial Intelligence

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ABSTRACT

Financial forecasting models are often evaluated by pooling observations from many assets, but strong pooled performance may reflect persistent differences between assets rather than changes through time. This study trained a Transformer on 60-session windows from 79 instruments across six asset families to predict an adverse event during the next ten sessions. Chronological splits with explicit label-overlap controls, train-only preprocessing and validation-only calibration limited information leakage. The Transformer appeared strong when all test windows were pooled (ROC-AUC 0.790), yet a training-only asset-prevalence baseline scored 0.824 and the Transformer's within-asset ROC-AUC was 0.492. Removing or swapping identity changed predictions, whereas reversing, permuting or shifting temporal order barely changed pooled rankings. No learned model met the project-internal gate for robust ordered temporal skill. Controlled simulation then showed that heterogeneous asset priors alone can create high pooled discrimination and that the diagnostics recover deliberately planted temporal signal. The contribution is therefore a simulation-validated diagnostic framework for separating cross-sectional shortcut learning from genuine temporal forecasting skill in multi-asset models, rather than evidence for a deployable forecaster.

Keywords: financial time series; Transformer; shortcut learning; pooled evaluation; within-asset ROC-AUC; model interpretability.

1. Introduction

Pooling financial series lets one shared model learn from more observations and potentially capture patterns that recur across assets, rather than fitting separate models to each series. This is the attraction of global forecasting, but it also creates an evaluation trap. Suppose one asset has frequent adverse events and another has few. A score that is high for every observation of the first asset and low for every observation of the second can rank the pooled sample well without identifying a single high-risk date within either asset. A flexible model may learn this shortcut from an asset identifier, from scale or patterns of missing values, or from persistent feature levels. Its pooled metric can then look like forecasting skill through time even when the model makes little use of the chronological order of the 60-session history.

This distinction matters for financial machine learning. For a model intended to forecast changing risk through time, it must distinguish high- and low-risk periods within an asset rather than only rank assets by historical risk. Ranking different assets against one another, a cross-sectional objective, may still be useful for screening. The distinction also matters scientifically because global forecasting models, which fit one shared model across many related series, are increasingly used in time-series forecasting. The benefits of global estimation are well established, but pooled evaluation can mix differences between assets with the ability to identify changing risk within an asset (Salinas et al., 2020; Montero-Manso and Hyndman, 2021). The problem becomes acute when target prevalence differs across assets or families.

This dissertation asks: does a pooled multi-asset Transformer learn genuine temporal market dynamics, or does its apparent predictive performance arise mainly from static cross-sectional priors? The empirical task predicts whether an instrument will experience a large endpoint loss, a severe interim fall relative to its forecast-date price, or a volatility spike over the next ten observed sessions. The model sees 60 sessions of engineered price, return, volatility, trend, volume, drawdown and macroeconomic information. It is assessed chronologically on a broad panel of equities, bonds, commodities, foreign exchange, crypto assets and real-asset proxies.

The aim is to determine whether pooled multi-asset Transformer performance reflects time-varying market information or mainly stable differences among instruments. The objectives are to: (1) build and chronologically evaluate a leakage-aware forecasting framework; (2) separate between-asset ranking from within-asset timing using static priors and tests that alter asset identity or chronological order; and (3) test whether the diagnosis is specific to the Transformer through a deliberately limited five-model comparison and controlled simulation.

These objectives produce three contributions. First, the study implements explicit future-horizon labels, strict within-asset windows, train-only preprocessing, chronological validation, controls for overlapping labels at split boundaries, and validation-only calibration. Second, it combines static asset and family priors, within-asset evaluation, identity removal and swapping, representation probing and tests that deliberately alter chronological order to determine what pooled discrimination actually measures. This is the primary empirical contribution. Third, it tests whether the diagnosis is model-specific through a deliberately limited five-model comparison and validates the diagnostic logic using 1,040 controlled simulations. The simulations show both failure and recovery: static prevalence heterogeneity can inflate pooled ROC-AUC without temporal signal, while planted dynamics raise within-asset performance and make predictions sensitive when chronological order is disrupted.

The practical relevance is primarily in financial model validation and risk governance. A strong pooled score may encourage analysts to treat a model as responsive to changing market conditions when it is mainly ranking instruments by persistent differences in historical risk. The framework developed here provides tests for distinguishing those cases before a model is considered for monitoring or decision support. The fitted Transformer itself did not pass the required evidence threshold for time-varying risk prediction, so no deployment or trading-performance claim is made.

2. Related Work

2.1 Global forecasting and entity identity

Global models estimate one forecasting function across many series rather than fitting one model per series. DeepAR shares an autoregressive recurrent network across related series to forecast full future distributions and can condition on series-level covariates (Salinas et al., 2020). Montero-Manso and Hyndman (2021) show more generally that a sufficiently expressive global model need not be more restrictive than a collection of local models; its parameter count also need not grow with the number of series. These results justify pooled estimation even for heterogeneous panels. Their usual evaluation target, however, is aggregate forecast error for future values. Such an error can establish useful prediction without answering whether a pooled binary classifier identifies changing high-risk periods within a series or mainly persistent differences between series.

Sirignano and Cont (2019) provide a closer financial comparison. They trained a universal recurrent model on billions of high-frequency order-book observations pooled across US equities to predict subsequent price-move direction. Performance transferred across time and to stocks excluded from training, while longer order-flow histories improved prediction. This suggests that a shared relationship between recent market information and subsequent price movements can generalise across stocks, providing a strong case for financial pooling. The present task differs in frequency, features and outcome: it classifies overlapping ten-session adverse events across six asset families whose event prevalences differ sharply. The present study does not test generalisation to previously unseen assets. A static prevalence control is consequently necessary before pooled AUC can be interpreted as temporal skill.

Asset identity can also provide legitimate predictive information rather than necessarily acting as a shortcut. The Spatial-Temporal Identity (STID) model attaches spatial and temporal identity embeddings to a deliberately simple multivariate forecaster to resolve indistinguishable samples, then evaluates aggregate forecasting error (Shao et al., 2022). The Market-Guided Stock Transformer (MASTER) learns temporal and cross-stock representations guided by market information for joint stock forecasting (Li et al., 2024). These studies demonstrate why asset identity or information from other assets may improve a valid prediction objective. They do not ask whether a constant entity prevalence can explain pooled classification, nor do they compare rankings before and after disrupting temporal order. This dissertation therefore does not treat identity as inherently problematic. It separates identity-conditioned ranking between assets from the ability to rank changing risk within the same asset.

2.2 Transformers and strong simple baselines

The Transformer replaces recurrence with self-attention over a sequence (Vaswani et al., 2017). Time-series variants alter how temporal positions, variables and static information enter that mechanism. Temporal Fusion Transformers combine recurrent local processing, attention, gating, variable selection and static covariates for multi-horizon probabilistic forecasting (Lim et al., 2021). MASTER instead focuses attention on interactions among stocks and market states. Both illustrate plausible routes by which attention can use temporal or cross-entity context. Neither architecture guarantees that a fitted score depends on chronological order, and attention weights alone do not establish the mechanism used.

Simple controls remain essential because architecture complexity is not evidence of temporal information. DLinear decomposes a series and applies simple linear mappings; it outperformed several elaborate Transformers on standard long-horizon benchmarks (Zeng et al., 2023). Those benchmarks concern future-value errors rather than heterogeneous event priors, but the result motivates a broader principle: the strongest simple comparator should match the claim being tested. Here that set includes not only a flattened logistic model, multilayer perceptron (MLP), long short-term memory network (LSTM) and temporal convolutional network (TCN), but also training-only asset and family prevalences. The deliberately limited five-model comparison tests whether the diagnosis recurs across model classes; it is not an architecture search.

2.3 Evaluation, shortcuts and interpretation

Time-series validation must preserve order because random splits can leak future information and misrepresent performance under non-stationarity (Cerqueira et al., 2020). Financial event labels add a separate overlap problem: neighbouring forecast origins can share future returns. Purging removes earlier labels whose future intervals cross an evaluation boundary, whereas embargoing adds a gap after that boundary (López de Prado, 2018). This study audits the actual interval endpoints rather than assuming that a nominal horizon-sized gap is sufficient across mixed trading calendars.

What the metric is measuring also matters. ROC-AUC is the probability that a randomly selected positive receives a higher score than a randomly selected negative. A pooled multi-asset AUC combines two kinds of pair: dates from the same asset and dates from different assets. If event prevalence differs by asset, many between-asset pairs can be ordered correctly by a score that never changes through time. Covariate-adjusted ROC analysis shows more generally that discrimination can change after conditioning on a covariate (Janes and Pepe, 2009). Here asset identity is the conditioning variable. Pair-weighted within-asset AUC removes between-asset pairs, while static priors quantify how much pooled ranking is available from training prevalence alone.

The broader machine-learning literature defines shortcut learning as reliance on predictive signals that satisfy a benchmark but not the intended mechanism (Geirhos et al., 2020). Representation probes test whether information such as asset identity can be decoded from a model's learned internal representation, although successful decoding alone does not prove that the model uses this information to make predictions (Hewitt and Liang, 2019). Attention weights should not be treated automatically as explanations (Jain and Wallace, 2019). Attribution results should also change when the model's learned parameters are randomised; otherwise they provide weak evidence that the explanation is specific to the trained model (Adebayo et al., 2018). The novelty claim is therefore deliberately narrow: this dissertation combines training-only static priors, within-asset metrics, identity ablation and probing, tests that alter chronological order while keeping the forecast endpoint fixed, cross-model comparison and controlled simulation to distinguish cross-sectional shortcut learning from temporal skill in a heterogeneous financial panel. It does not claim that any individual diagnostic is new. Table 1 locates that combined design against the closest comparators.

Table 1. Direct comparison with the closest methodological work.

Work	Relevant design	Gap addressed / remaining limitation
DeepAR; global/local theory	future-value forecasting with parameters shared across related series	aggregate errors do not isolate within-series event timing
Sirignano and Cont	pooled order books predict price-move direction, including unseen stocks	no heterogeneous long-horizon event-prior control
STID	explicit node/time identities improve multivariate forecast errors	no static-prior or within-asset ranking test
DLinear	linear mappings challenge Transformers on long-horizon errors	no pooled asset-prevalence problem
MASTER	market-guided Transformer models temporal and cross-stock structure	no training-prior or order-perturbation control
This dissertation	pooled event prediction plus within-asset, identity and order tests	historical test evidence is adaptive; independent confirmation remains open

The closest strands therefore answer complementary questions rather than the question posed here. Global-forecasting work asks whether shared estimation can improve future-value prediction; STID and MASTER ask how asset identity or cross-asset context can improve an aggregate objective; DLinear asks whether architectural complexity beats a strong simple predictor. Shortcut and conditional-ROC work instead ask what a successful metric is measuring. This dissertation joins those strands at the evaluation stage. It asks whether improvement in a pooled financial event metric survives a baseline based only on training-set asset prevalence, evaluation using only pairs of observations from the same asset, tests that remove or swap asset identity, and tests that disrupt chronological order. Its contribution is this combined set of falsification tests, whose diagnostic behaviour is validated in controlled simulation, not a claim that pooled models, identity embeddings, AUC conditioning or temporal perturbations are individually new.

The comparison also separates transfer from timing. DeepAR and the global/local framework justify estimating common parameters when related series share structure, but their usual aggregate forecast-error objective can improve even if the model mostly learns persistent differences in scale or seasonality. Sirignano and Cont offer stronger evidence of transferable financial dynamics because one model generalised across order books and to stocks excluded from training, with additional order-flow history improving prediction. Their outcome, frequency and entity support differ from the present ten-session event task, however, so that result does not remove the need for a prevalence-only control here. Conversely, STID makes identity an intentional predictive input and MASTER uses market-wide context; both are legitimate designs when the evaluation objective rewards differences or relationships across assets. The issue is not whether identity is present, but whether the reported metric distinguishes its static contribution from changing state.

DLinear warns that a sophisticated sequence model should not be credited when a simpler model matches the relevant objective. Here that principle applies to information rather than parameter count: the strongest baseline is deliberately non-temporal. A learned model must outperform static group information, rank changing risk within an asset, and show that its predictions meaningfully depend on chronological order. Together, these tests address the methodological gap identified above.

Taken together, the closest literature suggests three distinctions before pooled performance is interpreted as temporal forecasting. Global estimation can be useful without establishing the ability to identify changing risk through time; asset identity can legitimately support ranking between assets while still acting as a shortcut if the intended claim is timing risk within the same asset; and model complexity should be judged against baselines that contain the same relevant information, not only against simpler model architectures. These distinctions motivate the design here: static priors measure ranking available from persistent group differences, within-asset AUC removes between-asset pairs, and tests that alter chronological order assess whether the sequence itself contributes information beyond the engineered summary features.

3. Methodology

3.1 Data, forecast origin and inputs

The configured daily universe contained 80 instruments from six families: 39 equities, 11 bonds, eight commodities, six foreign-exchange instruments, 13 crypto assets and three real-asset proxies. Seventy-nine instruments formed valid windows in the corrected final split. The panel spans 4 January 2010 to 23 June 2026 where history is available and contains 306,174 observed open-high-low-close-volume (OHLCV) rows. Calendar-aligned placeholders were excluded from model windows; ETF weekend prices were not forward-filled to imitate the continuous crypto calendar. This avoids treating stale prices as observations.

Yahoo Finance supplied open, high, low, close, adjusted close and volume. Adjusted close was preferred for returns and the target because it accounts for distributions and splits; all 80 configured series had complete adjusted-close values for target construction, so the row-level close fallback was not required. Raw open, high, low and close were retained only to construct intraday range, open-close and overnight-gap variables. Raw price levels did not enter the network directly.

At each observed close t, the model received 60 within-asset sessions ending at t:

$$X_{i,t} = \left[x_{i,t-59}, \dots, x_{i,t}\right], \; where \; L = 60.$$

Each x contained 34 numerical channels scaled using training data only: 27 engineered market channels and seven macro/context channels. The market channels covered six price/return variables, six volatility variables, six momentum/trend variables, four volume/activity variables and five stress/drawdown variables. The seven context channels were the federal funds rate, 2-year and 10-year Treasury yields, the 10y-2y slope, the VIX equity-volatility index, the US high-yield spread and a broad dollar index. Federal Reserve Economic Data (FRED) observations were lagged by at least one business day before alignment. A separate 12-dimensional learned asset embedding was repeated at each timestep, giving the Transformer 46 channels after concatenation. No family label or explicit missingness indicator was supplied. Table 2 enumerates every direct input.

Table 2. Direct numerical channels supplied at each of the 60 timesteps.

Group	Direct channels
Price/return (6)	close return; log return; open-close return; high-low range; overnight gap; intraday range proxy
Volatility (6)	5-, 10- and 20-session volatility; downside volatility; rolling high-low range; EWMA volatility
Momentum/trend (6)	5- and 20-session cumulative return; moving-average distance; 5/20 crossover; RSI; MACD
Volume/activity (4)	volume change; 20-session volume z-score; volume/average ratio; price-volume interaction
Stress/drawdown (5)	60-session drawdown; distance from 20- and 60-session highs; negative-return streak; downside indicator
Macro/context (7)	DFF; DGS2; DGS10; T10Y2Y; VIXCLS; BAMLH0A0HYM2; DTWEXBGS

Note: *The public repository provides the exact instrument universe, provider symbols and acquisition code; raw provider data are not redistributed. Finance-specific indicators and macro series are defined in Appendix D.*

For each asset, a median imputer and standard scaler were fitted only on its training dates, then applied unchanged to validation and test dates. A feature with no observed training values became zero after scaling. There was no backward filling from future observations. Windows never crossed assets or split boundaries. This design answers the forecast-origin question directly: the Transformer saw engineered information available by asset i's close at t, together with an identity embedding, and nothing from t+1 onward. Figure 1 summarises the forecast construction and corrected chronological split.

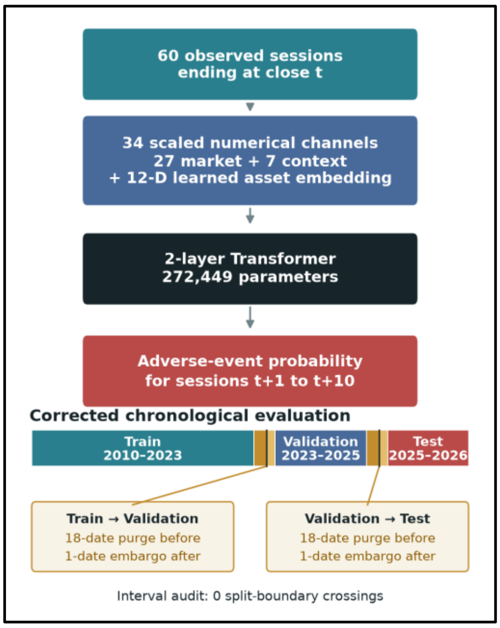

Figure 1. *Forecast design and chronological evaluation. The model receives 60 observed sessions through close t; only sessions t+1 to t+10 define the adverse-event label.*

3.2 Ten-session adverse-event target

Let $P_{i,t}$ be adjusted close and $r_{i,t} = \log[P_{i,t}/P_{i,t-1}]$. For horizon $H = 10$, the implementation computes:

$$R^{(10)}_{i,t} = P_{i,t+10}/P_{i,t} - 1,$$

$$D^{(10)}_{i,t} = min_{1\le k\le 10}\{P_{i,t+k}/P_{i,t} - 1\},$$

$$V^{future}_{i,t} = \sqrt{\textstyle\sum_{k=1}^{10} r^2_{i,t+k}},\ and\ V^{hist}_{i,t} = \sqrt{10}sd[r_{i,t-19:t}],$$

where sd is the rolling sample standard deviation (pandas ddof=1).

The binary label is:

$$y_{i,t} = \mathbf{1}\{R^{(10)}_{i,t} \le -0.05 \vee D^{(10)}_{i,t} \le -0.07$$
$$\vee V^{future}_{i,t} \ge 2V^{hist}_{i,t}\}.$$

Thus, y = 1 if at least one of three events occurs after t: a 5% loss by session t+10; a 7% maximum adverse excursion (the largest fall relative to the forecast-date price P(i,t)) during the next ten sessions; or realised volatility at least twice its trailing 10-session-scaled sample-volatility benchmark. Only t+1 through t+10 enter the label. This composite target captures adverse conditions, but it is not equally rare or equally severe across families. Validation prevalence ranged from 2.10% for bonds to 49.56% for crypto. That heterogeneity is central to the later falsification and limits cross-family interpretation.

3.3 Chronological split and leakage controls

The corrected fold used train dates from 4 January 2010 to 24 November 2023, validation dates from 14 December 2023 to 23 February 2025, and test dates from 15 March 2025 to 13 June 2026. Strict windowing produced 245,055 training, 20,494 validation and 21,514 test examples. Validation data determined training duration, calibration and the decision threshold. Test labels were not used for any of these choices.

Neighbouring ten-session labels can share future returns. Purging removes observations from the earlier split when their label intervals extend across the next split boundary, while embargoing removes the first date after that boundary to reduce residual dependence. A nominal ten-date purge on the pooled calendar still left 120 training and 60 validation labels crossing the boundaries. The final correction therefore removed 18 pooled-calendar dates before each boundary and embargoed one date after it. An explicit audit of all label intervals then found zero crossings. The longer gap was necessary because assets follow different observed-session calendars.

3.4 Transformer and training protocol

The 46-channel sequence was mapped to 128 dimensions and combined with fixed sinusoidal positional encodings. The encoder had two pre-normalised layers, four attention heads, a 256-unit feed-forward block, GELU activation and dropout 0.3. Temporal-attention pooling reduced the 60 hidden states to one vector, followed by layer normalisation, dropout and a linear logit head. The complete model had 272,449 parameters.

Training used a differentiable soft-F1 loss, AdamW with learning rate 3 x 10^-4 and weight decay 10^-4, batch size 1,024, gradient clipping and at most 12 epochs. Early stopping used a patience of three epochs. Seeds 7, 42 and 123 produced a probability ensemble. Calibration method and the F1 decision threshold were selected on validation only. The model was trained with mixed precision where supported.

Ranking and probability metrics use different score objects. Raw ensemble scores were used for ROC-AUC, precision-recall AUC (PR-AUC) and within-asset ROC-AUC because isotonic calibration can create ties and therefore change ranking metrics. Validation-fitted isotonic probabilities were used for Brier score, log loss and threshold-based F1 and balanced accuracy. This separation prevents calibration-induced ties from being mistaken for changes in ranking performance.

3.5 Metrics, baselines and falsification tests

Pooled ROC-AUC measures threshold-independent ranking across all test windows, but can reward between-asset prevalence differences. Within-asset ROC-AUC asks whether high-risk dates rank above low-risk dates for the same asset. For each asset with both classes, AUC_i was weighted by its number of positive-negative pairs:

$$AUC_{within} = (\Sigma_i n_{i,+} \cdot n_{i,-} \cdot AUC_i) \,/\, (\Sigma_i n_{i,+} \cdot n_{i,-}).$$

This pair-weighted statistic uses the same pair interpretation as pooled AUC while removing between-asset pairs. Per-asset macro AUC gives every eligible asset equal weight. PR-AUC is reported because positives are imbalanced. Balanced accuracy weights sensitivity and specificity equally; F1 combines precision and recall at the validation-selected threshold. Brier score and log loss assess probability accuracy, with log loss penalising confident errors more strongly.

These metrics answer different questions and are not interchangeable. Pair weighting estimates discrimination over all eligible same-asset positive-negative pairs, so assets with greater class support contribute more. Macro AUC asks whether performance is broad by giving each eligible asset equal weight, but becomes noisier for assets with few events. Pooled AUC mixes ranking between assets with ranking through time. Reporting all three prevents the conclusion from depending on a single aggregation choice. Threshold-based metrics and calibration still describe classification and probability quality, but they cannot show whether a score varies meaningfully through time. For that reason, the static-prior and within-asset comparisons are interpreted before the threshold results.

The training-only asset prior assigns every validation or test date for an asset the same prevalence estimated from that asset's training labels. The family prior does the same at the coarser family level. Because these baselines are constant through time within an asset or family, any pooled discrimination they achieve arises from cross-sectional differences in training prevalence rather than temporal ranking. The asset prior's within-asset AUC is therefore 0.5 by construction whenever both classes are present. The Transformer without asset ID removes the explicit asset embedding. A cyclic ID swap measures prediction sensitivity to incorrect identities, while a regularised representation probe tests whether asset identity can be decoded from the model's learned internal representation.

Three tests deliberately disrupt chronological order without changing the asset or final forecast origin: reverse the 60 sessions, apply one deterministic permutation, or circularly shift the sequence. A genuinely order-dependent model should change its rankings materially. These tests assess reliance on ordered observations, not whether every input is time-varying: rolling volatility, momentum, trend and drawdown channels already summarise parts of the historical path. A project-internal gate, frozen before the final model comparison, required evidence from paired within-asset improvement, sensitivity to chronological order and dependence-aware uncertainty. It was applied consistently to the Transformer, MLP, LSTM, TCN and flattened logistic model.

The gate deliberately separates three questions that a single point estimate cannot answer. First, is conditional ranking above chance with dependence-aware uncertainty? Second, is any lift stable across seeds and paired observations rather than driven by composition? Third, does performance deteriorate when the chronological information claimed to matter is disrupted? A model was considered to show robust ordered temporal skill only when all three questions supported the same interpretation. This criterion is stricter than reporting an ensemble AUC above 0.5, but the simulation below tests whether it still permits recovery when ordered signal truly exists.

Finally, a controlled simulation varied asset-prior heterogeneity, dynamic signal strength and persistence. It compared pooled and within-asset AUC and repeated temporal perturbations over 20 seeds per condition. The complete simulation design comprised 1,040 runs, all of which completed successfully. Simulation provides a known data-generating process: it tests whether the diagnostics remain at chance without dynamics and respond when ordered signal is deliberately planted.

4. Results

4.1 Apparent pooled performance was largely cross-sectional

The asset-conditioned Transformer achieved raw pooled ROC-AUC 0.789814 and PR-AUC 0.421056 over 21,514 test windows. Its validation-selected probabilities gave Brier score 0.110920, log loss 0.360548, F1 0.511420 and balanced accuracy 0.745220. Considered alone, those values suggest useful predictive performance.

The static controls changed that interpretation. The training-only asset prior achieved pooled ROC-AUC 0.823906 and the family prior 0.816549. Both exceeded the Transformer despite being constant through time within each group. The asset prior is a particularly strong falsification control: it assigns one training-period prevalence to every date of the same asset, so its within-asset ROC-AUC is exactly 0.500000 by construction. Its pooled AUC of 0.823906 therefore arises entirely from cross-sectional differences in event prevalence, not from identifying changing risk dates.

The reason is visible from the pair interpretation of AUC. Pooled AUC compares every positive test window with every negative test window. Some comparisons involve two dates from the same asset, but many involve different assets. When, for example, positives occur much more often in one family than another, a constant asset score can order many between-asset pairs correctly. Within-asset AUC discards those between-asset comparisons and asks the intended timing question only: within a fixed instrument, did the model score its adverse dates above its non-adverse dates? The Transformer's pair-weighted within-asset AUC was 0.491638, effectively chance, while its equal-asset macro AUC was 0.538743. Overall, the Transformer distinguished historically high-prevalence assets from low-prevalence assets far better than it identified adverse periods within a given asset. Figure 2 makes this contrast explicit.

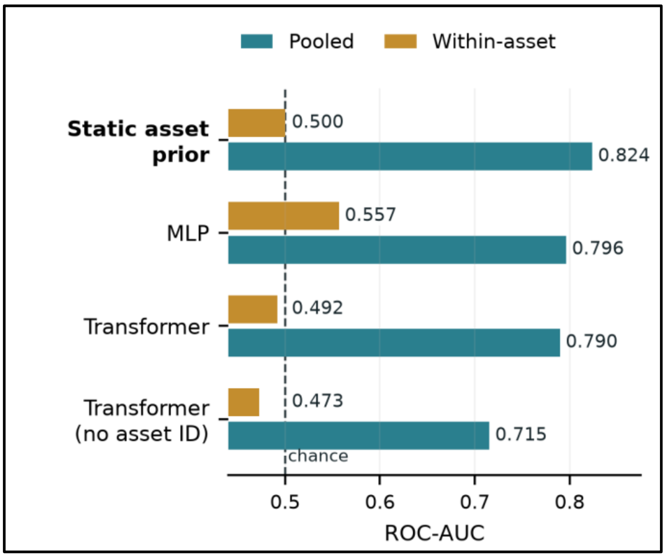

Figure 2. *Pooled versus pair-weighted within-asset ROC-AUC. The static asset prior scores 0.824 pooled but 0.500 within-asset because its score never changes through time.*

Table 3 shows that the pattern was not specific to one architecture. The MLP had the strongest learned pooled AUC (0.796478) and within-asset estimate, 0.556967 with dependence-aware interval [0.504831, 0.600824]. This supports a modest within-asset predictive association, but not by itself robust ordered temporal skill. The project-internal gate also required consistent improvement across seeds and paired observations, together with a meaningful performance loss when chronological order was disrupted. Those requirements were not satisfied. Other models either had uncertainty intervals crossing 0.5 or similarly showed insufficient sensitivity to chronological order. No learned model passed the complete gate: 0/5. This does not make temporal prediction impossible; it distinguishes conditional association from ordered forecasting.

Table 3. Cross-model ranking on the corrected historical test split.

Model	Pooled ROC-AUC	Within-asset ROC-AUC
Static asset prior	0.823906	0.500000
MLP	0.796478	0.556967
Transformer encoder	0.789814	0.491638
TCN	0.775881	0.547453
LSTM	0.692870	0.518344
Flattened logistic	0.576220	0.512171

The MLP illustrates why all parts of the gate must be satisfied together. Its interval supports modest conditional discrimination; chronology tests ask whether that association requires the ordered 60-session path. Failure of the chronology gate means that the association could not be attributed robustly to that path, not that all changing-state information was absent: engineered rolling features already encode time-varying summaries. Conversely, an order-sensitive model with chance-level within-asset AUC is also insufficient. Robust temporal forecasting therefore requires both within-asset discrimination and evidence that performance depends meaningfully on chronological order. None of the five models supplied that combination.

The MLP therefore provides the strongest learned within-asset ranking result and merits independent replication. Its interval quantifies uncertainty around the within-asset AUC, but does not establish consistent paired improvement or dependence on chronological order. A secondary Transformer retrained with a within-asset objective was also numerically interesting at 0.561924, but its interval crossed 0.5 and its uncertainty, paired-improvement and chronology requirements were not met. Neither result meets the project-internal gate for robust temporal forecasting.

4.2 Identity mattered; chronological order barely did

Removing the 12-dimensional asset embedding reduced pooled AUC from 0.789814 to 0.715477 and did not improve within-asset AUC, which fell to 0.472570. This ablation isolates the explicit identifier, but not all implicit identity. Persistent volatility, trading calendar, volume scale and feature missingness can still reveal which asset generated a window. The decline therefore shows that explicit identity contributed to pooled ranking; the absence of within-asset recovery shows that removing it did not expose a hidden temporal forecaster.

Cyclically swapping asset IDs changed predicted probabilities by a mean absolute amount of 0.090025 and reduced pooled AUC to 0.682928 while leaving each market-history window fixed. This controlled mismatch demonstrates score sensitivity to the supplied identity, although it does not establish that identity was the model's only shortcut. A representation probe decoded asset identity from the model's learned internal representation with test accuracy 0.166930, more than thirteen times the approximate 1/79 = 0.0127 chance rate. Successful decoding shows that asset identity is represented internally, but does not by itself prove that the model uses that information to make predictions; the ablation and ID-swap tests provide the complementary behavioural evidence. Together the three diagnostics support identity dependence without converting that dependence into a causal market explanation.

Disrupting chronological order produced a very different response. Reversal, deterministic permutation and circular shift yielded pooled AUCs 0.791263, 0.788585 and 0.790321. Each was within 0.0015 of the original unperturbed pooled AUC. Reversal changes the direction of the sequence, deterministic permutation consistently breaks local ordering, and circular shifting changes where the sequence begins while retaining the same observations. These transformations keep the asset, forecast target and underlying window observations fixed while altering their order in different ways. Engineered rolling channels still summarise parts of the recent history. The small changes therefore do not prove that no time-varying information existed. They do, however, weaken the claim that the original ordered 60-session path drove the observed pooled ranking.

Feature-group occlusion, masking selected groups of inputs, reinforced the identity diagnosis but was not treated as a faithful explanation. Removing macro/context inputs produced mean prediction displacement 0.160730; masking days 41-60 produced 0.141292; removing the asset embedding produced

0.136454; and masking returns/momentum produced 0.129481. Following model-randomisation sanity-check logic (Adebayo et al., 2018), the trained-versus-randomised attribution rank check failed its threshold. These values are therefore sensitivity diagnostics, not evidence that macro variables were causally or definitively the most important features. None of the 14 predefined regimes passed the temporal-dynamics gate, so the evidence did not support a regime-specific interpretation of the model's internal states.

4.3 Controlled simulation validated the diagnostic mechanism

The simulation created two controlled worlds in which the data-generating mechanism was known by construction. World A contained heterogeneous asset event priors but no dynamic signal. As prior heterogeneity increased, static asset-prior pooled AUC rose by 0.400021 (95% interval 0.391244 to 0.408797). In the core persistent setting it moved from approximately 0.499 to 0.899, even though the classifier's within-asset AUC remained 0.500086 (0.497032 to 0.503141). This reproduces the empirical concern in isolation: cross-sectional prevalence differences are sufficient to create apparently excellent pooled discrimination.

World B removed prior heterogeneity and planted an ordered, persistent temporal signal. Under strong signal, within-asset AUC rose to 0.783997 (0.781142 to 0.786852). Reversing the sequence then reduced AUC by 0.380991, while deterministic permutation reduced it by 0.102297. All simulation mechanism checks passed. These are not estimates of real market effect sizes; they test the behaviour of the diagnostics under controlled conditions.

This distinction matters for the empirical interpretation. A weak real-data response to temporal perturbation could otherwise mean that the tests were too severe, badly designed or simply unable to detect sequence dependence. The simulation shows that the diagnostics can distinguish these cases within the tested mechanism: they remained at chance when only static heterogeneity was present, but recovered within-asset skill and sensitivity to chronological order when temporal signal was deliberately planted. Figure 3 displays both worlds and links the high empirical pooled AUC to a concrete failure mechanism rather than to a visual analogy.

Simulation does not prove that real markets follow this synthetic process, nor that the simulated effect sizes transfer to finance. Its role is narrower: to test whether the proposed mechanism is sufficient to reproduce the observed metric pattern and whether the diagnostic suite behaves as intended when the underlying mechanism is known. Prior heterogeneity alone reproduced the qualitative pattern observed empirically, while the same evaluation framework did not automatically reject a model when deliberately planted ordered signal was present. The empirical and simulated evidence therefore play different roles: the market data motivate the shortcut diagnosis, while the simulation shows that the proposed mechanism can generate that pattern and that planted temporal signal would have produced a different diagnostic response.

The two panels should therefore be read as a controlled contrast, not as two forecasts of market performance. In panel A, increasing heterogeneity changes only which assets tend to be positive; it does not create useful ordering of dates within an asset. The widening gap between pooled prior AUC and within-asset AUC is the shortcut mechanism. In panel B, prior heterogeneity is removed and ordered signal is increased. Within-asset discrimination then rises, and disrupting chronological order causes increasingly large performance losses. Improvement on both measures matters because it shows that the diagnostics distinguish the two planted mechanisms rather than simply penalising pooled models by design.

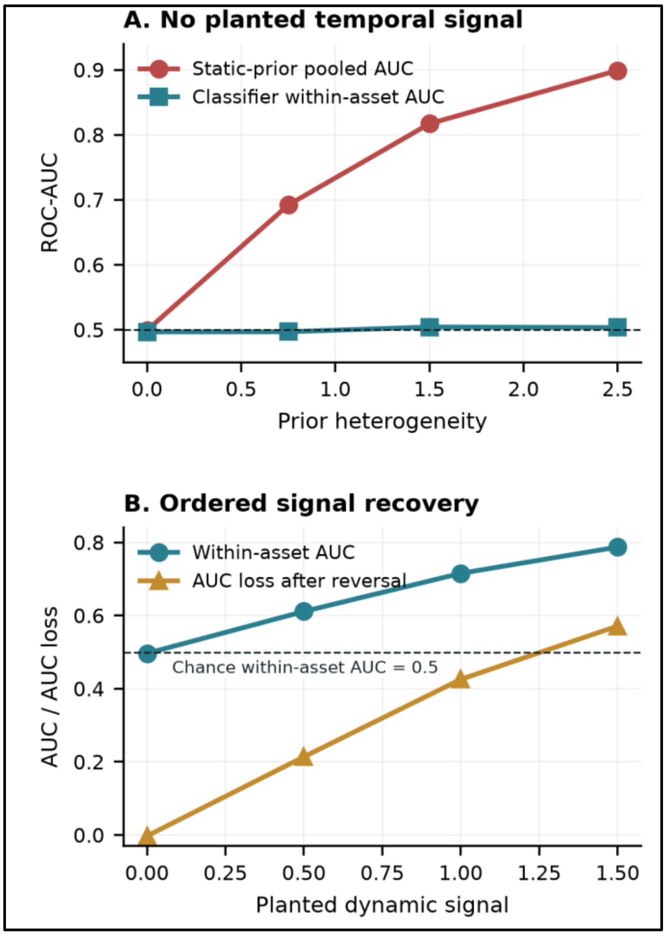

Figure 3. *Controlled mechanism checks. Static prior heterogeneity raises pooled AUC without within-asset timing skill (A); planted ordered signal raises within-asset AUC and increases the performance loss caused by reversing the sequence (B).*

4.4 Recovery checks did not overturn the finding

Three targeted redesigns tested plausible alternative explanations for the negative timing result. The first separated the static prevalence component from a learned residual intended to capture time-varying information. The second aligned part of the training objective directly with within-asset ranking. The third replaced the binary composite event with a continuous downside outcome. Table 4 reports the validation-selected historical test results.

Table 4. Bounded recovery tests on the corrected historical test split.

Recovery test	Main result	Interpretation
Static + dynamic residual	within AUC 0.525360; interval [0.443706, 0.602033]	paired-improvement and chronology checks failed
Within-asset objective Transformer	within AUC 0.561924; interval [0.492527, 0.625022]	uncertainty, paired-improvement and chronology checks failed
Continuous downside Transformer	equal-asset MAE 0.037816	asset mean 0.020896 and ridge 0.020691 were stronger

The residual model asks whether the strong static component had hidden a smaller time-varying signal; its interval crossed chance and the complete gate failed. The within-asset objective addresses the mismatch between pooled training and within-asset evaluation directly, but its interval also crossed 0.5 and neither paired improvement nor sensitivity to chronological order supported a robust temporal-skill conclusion. The continuous target tests whether defining the outcome as a binary event caused the failure, yet the simple training-mean and ridge baselines achieved substantially lower error. These are secondary robustness checks rather than additional headline experiments. Together they show that static/dynamic decomposition, objective alignment and target continuity did not overturn the main conclusion under the bounded designs.

Each recovery test removes one plausible explanation: the residual model separates static prevalence from the remaining learned signal, within-asset optimisation aligns training with conditional ranking, and the continuous downside target removes the binary event threshold. Their failures are informative but bounded. They reject these implementations on this opened historical period, not every possible decomposition, ranking loss or continuous outcome, and they are not independent confirmation.

5. Discussion and Conclusion

The central finding is a measurement result. A pooled Transformer ROC-AUC near 0.79 did not establish temporal forecasting skill. The stronger static prior, chance-level within-asset AUC, identity sensitivity and lack of material order sensitivity show that the pooled score mainly rewarded persistent cross-sectional structure. Target prevalence varied sharply across families, so between-asset positive-negative pairs dominated a pooled metric that appeared dynamic but could be ranked from training prevalence alone. Under known data-generating conditions, the controlled simulation reproduced this qualitative pattern and responded when ordered information was deliberately planted. It is a mechanism test, not external market replication or an estimate of real-market effect sizes.

This does not imply that global forecasting, identity embeddings or Transformers are invalid. A series identifier can calibrate persistent scale, seasonality or baseline risk, while pooled estimation can transfer genuinely shared structure. The problem is interpretive: identity conditioning becomes a shortcut relative to a temporal claim when the evaluation rewards stable asset differences and those differences are reported as changing-state skill. Cross-sectional ranking may support screening when the objective is to compare unconditional risk across assets; pooled AUC and identity are then relevant. If the objective is to detect changing risk within an asset, static-prior comparison, conditional ranking and temporal perturbation are indispensable. In this study the second objective was intended, and it was not supported.

The title is treated as an empirical question rather than a presupposed positive result. The project successfully implemented and interrogated a Transformer-based financial forecasting framework, but the evidence did not establish robust emergent temporal dynamics. Interpretability is partial: ablations, probes and perturbations reveal model dependence, but the failed attribution randomisation check prevents treating saliency or attention as faithful explanation. What emerged from the pooled benchmark was largely static cross-sectional structure rather than robust temporal dynamics.

Four findings should remain distinct. Pooled discrimination existed, but the static prior outperformed the Transformer and within-asset ranking was near chance, indicating a mainly cross-sectional effect. Identity and order interventions showed model dependence, not economic causality. Controlled simulation reproduced the proposed shortcut mechanism under known conditions but does not provide external market replication. The conclusion is therefore methodological: it concerns how pooled benchmarks should be evaluated, not a deployable temporal-risk model, a causal account of market stress, or proof that Transformers cannot learn financial dynamics.

The study has important limits. The historical test set became adaptive as successive falsification and recovery analyses were performed, so the empirical evidence is historical test evidence rather than untouched confirmation. The current-symbol universe may suffer survivorship bias because it does not fully reconstruct delisted instruments. The composite target is not rarity- or severity-equivalent across families; that heterogeneity is diagnosed, not removed. FRED inputs were lagged before alignment but were obtained as current-vintage data rather than fully point-in-time historical releases, and one credit-spread series had substantial missingness during the training period. Overlapping labels and common market shocks reduce effective support despite boundary-purge correction. Within-asset AUC removes between-asset pairs but does not by itself remove common-date dependence. The identity probe and occlusion tests establish recoverability and sensitivity, not causality. Finally, no externally confirmed positive temporal signal was found.

External validity is also limited: the panel uses one provider, one engineered feature set, one composite horizon and one historical era. Five models strengthen method generality but do not exhaust architectures; further test-driven searching would deepen adaptivity. Simulation validates diagnostic logic under a known mechanism, not market realism. These boundaries preserve the methodological result while precluding a universal statement about financial predictability.

Practically, the framework is useful for financial model validation rather than deployment. The fitted Transformer is not validated for operational market-risk monitoring, and no trading or profitability claim follows. Before relying on a strong pooled score, practitioners can compare training-only group priors, within-asset and equal-asset metrics, identity removal or swapping, and sensitivity to chronological order, then verify diagnostic behaviour in controlled simulation. This helps distinguish persistent cross-sectional ranking from genuine responsiveness to changing conditions.

6. Future work

Future work should prioritise independent confirmation rather than further tuning on the opened test set. A preregistered replication using another provider or a prospective period should freeze the target, features, candidate models, priors, within-asset metric and stopping rule before outcomes are inspected. Leave-one-asset/family-out tests could assess transfer, while asset-relative targets test whether cross-asset information improves within-asset timing.

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Appendix A. Extended empirical diagnostics

This appendix provides supplementary evidence for the main interpretation without changing the evaluation rules used in the body. Figure A1 compares pooled and pair-weighted within-asset discrimination across models. Figure A2 compares interventions on asset identity with changes to chronological order for the Transformer. Table A1 reports the main identity and sensitivity diagnostics.

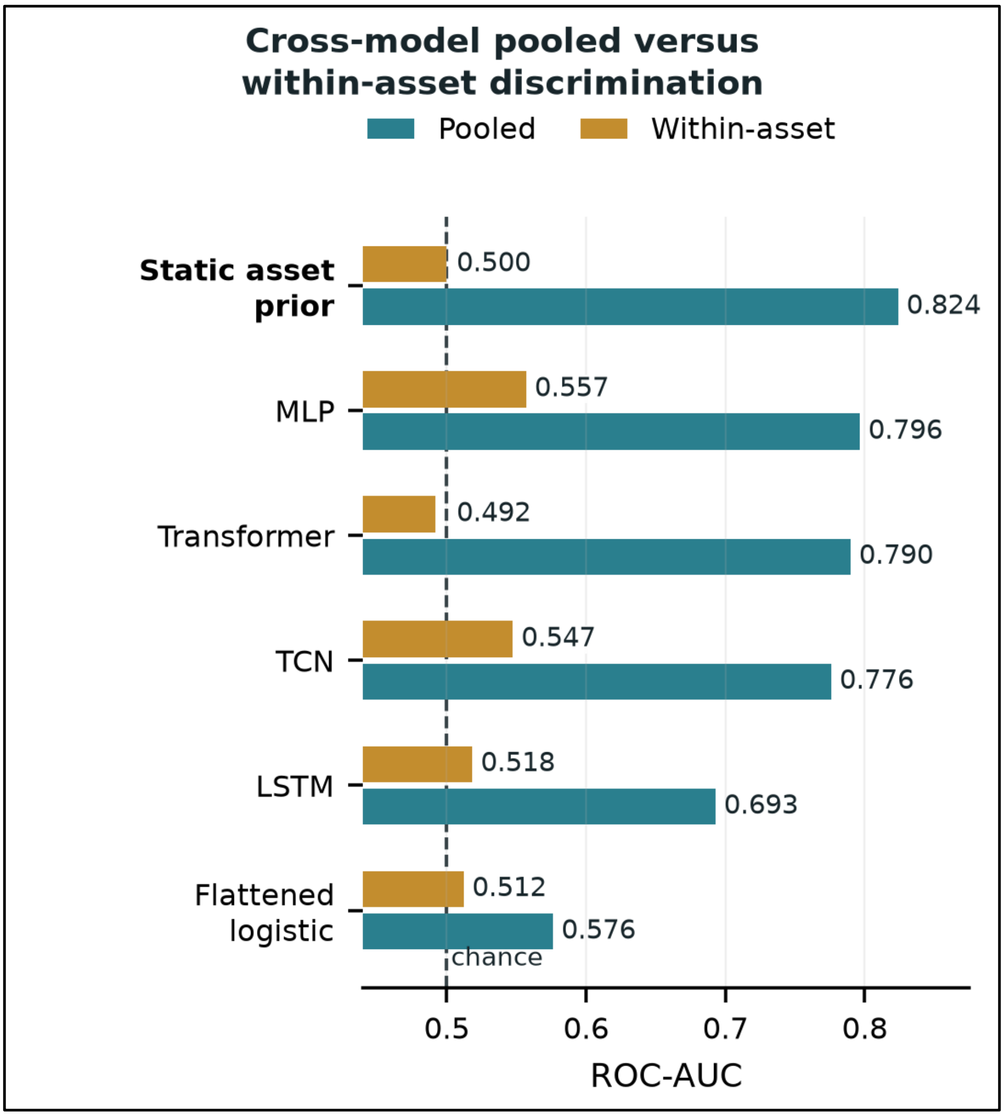

Figure A1. *Cross-model pooled versus pair-weighted within-asset ROC-AUC. The static asset prior shows how strong pooled ranking can arise without within-asset timing.*

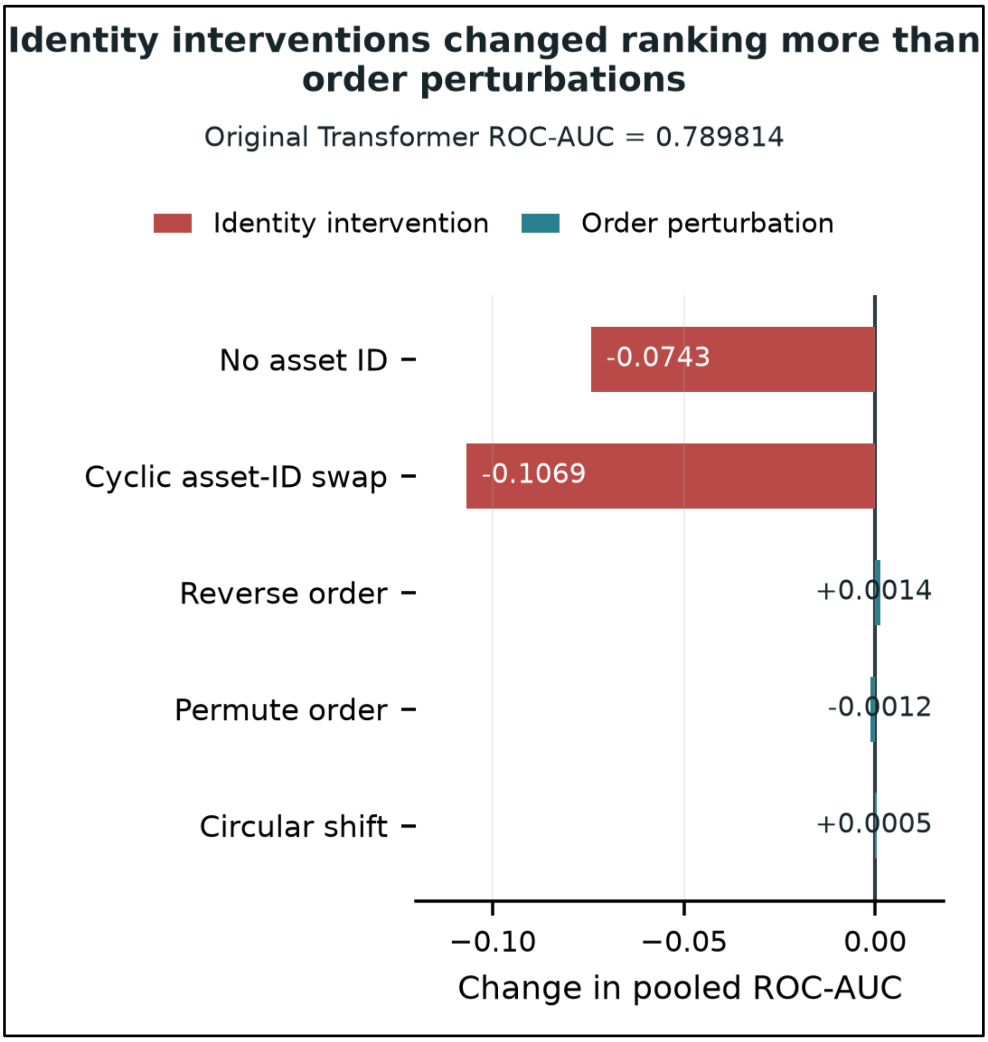

Figure A2. *Change in pooled ROC-AUC relative to the original Transformer. Removing or swapping asset identity changes ranking materially more than reversing, deterministically permuting or circularly shifting the same windows. These are behavioural sensitivity tests, not causal attributions.*

Table A1. Supplementary identity and interpretability diagnostics

Diagnostic	Value	Interpretation
Transformer without asset ID: pooled ROC-AUC	0.715477	Removing the explicit asset embedding materially reduced pooled ranking.
Transformer without asset ID: within-asset ROC-AUC	0.472570	Removing the asset embedding did not recover within-asset timing.
Cyclic asset-ID swap mean $\|\Delta p\|$	0.090025	Predicted probabilities were sensitive to the supplied asset identity.
Cyclic asset-ID swap pooled ROC-AUC	0.682928	Supplying incorrect asset identity materially degraded pooled ranking.
Asset probe accuracy	0.166930	Asset identity remained decodable from the learned representation; chance ≈ 0.0127.
Macro/context occlusion displacement	0.160730	Sensitivity diagnostic only; not a causal attribution.
Days 41–60 occlusion displacement	0.141292	Sensitivity diagnostic only; not a causal attribution.
Asset-embedding occlusion displacement	0.136454	Sensitivity diagnostic only; not a causal attribution.
Returns/momentum occlusion displacement	0.129481	Sensitivity diagnostic only; not a causal attribution.

A.1 Target prevalence structure

Because pooled ROC-AUC can reward persistent differences between assets as well as changes through time, the target prevalence structure is shown directly. Figures A3 and A4 use the same 79-asset model-evaluation endpoint population as the reported forecasting results: only forecast origins with complete 60-session lookbacks and valid target horizons are included. Figure A3 compares positive-label prevalence across the six asset families in the training, validation and test splits. Figure A4 compares each asset's training prevalence with its test prevalence and colours points by family. These plots are descriptive diagnostics: they show the extent and persistence of baseline-risk heterogeneity but do not establish its cause.

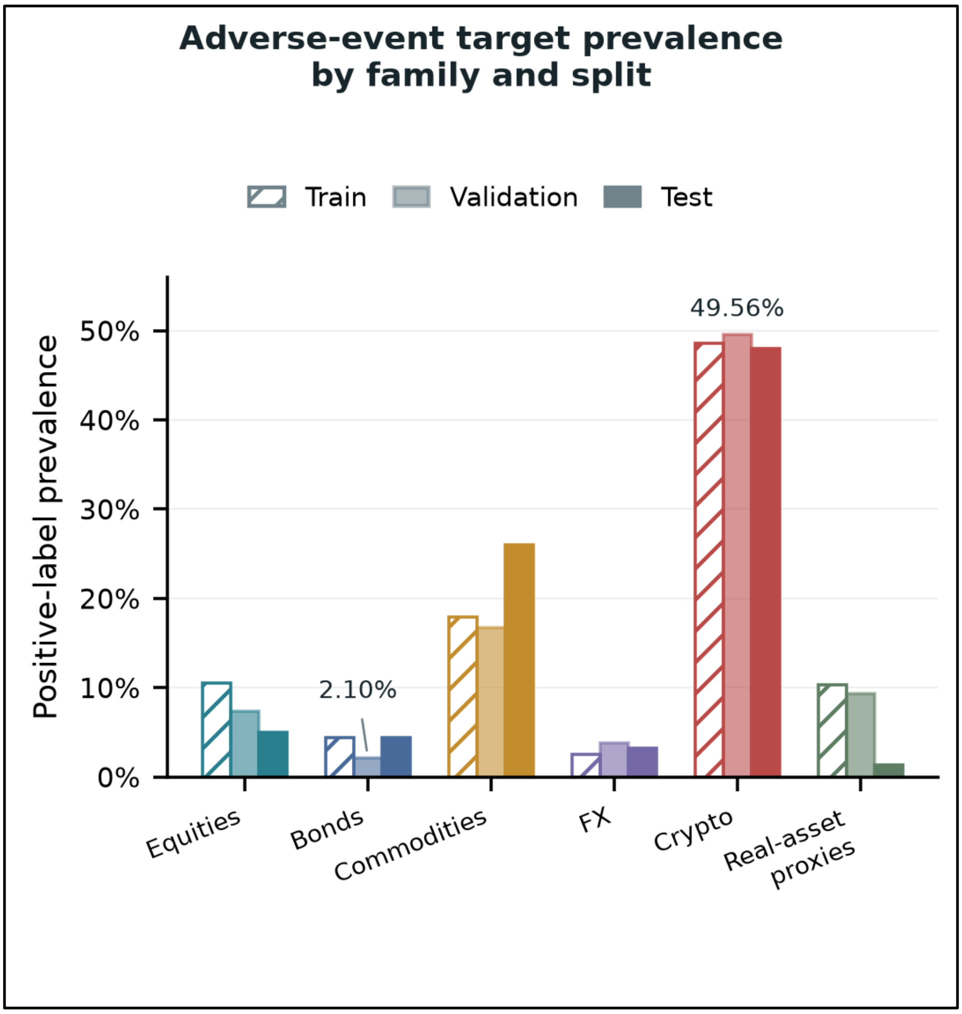

Figure A3. *Target prevalence by asset family and split for the 79-asset model-evaluation endpoint population. The composite event definition produces materially different positive-label rates across families; in validation, prevalence ranges from 2.10% for bonds to 49.56% for crypto.*

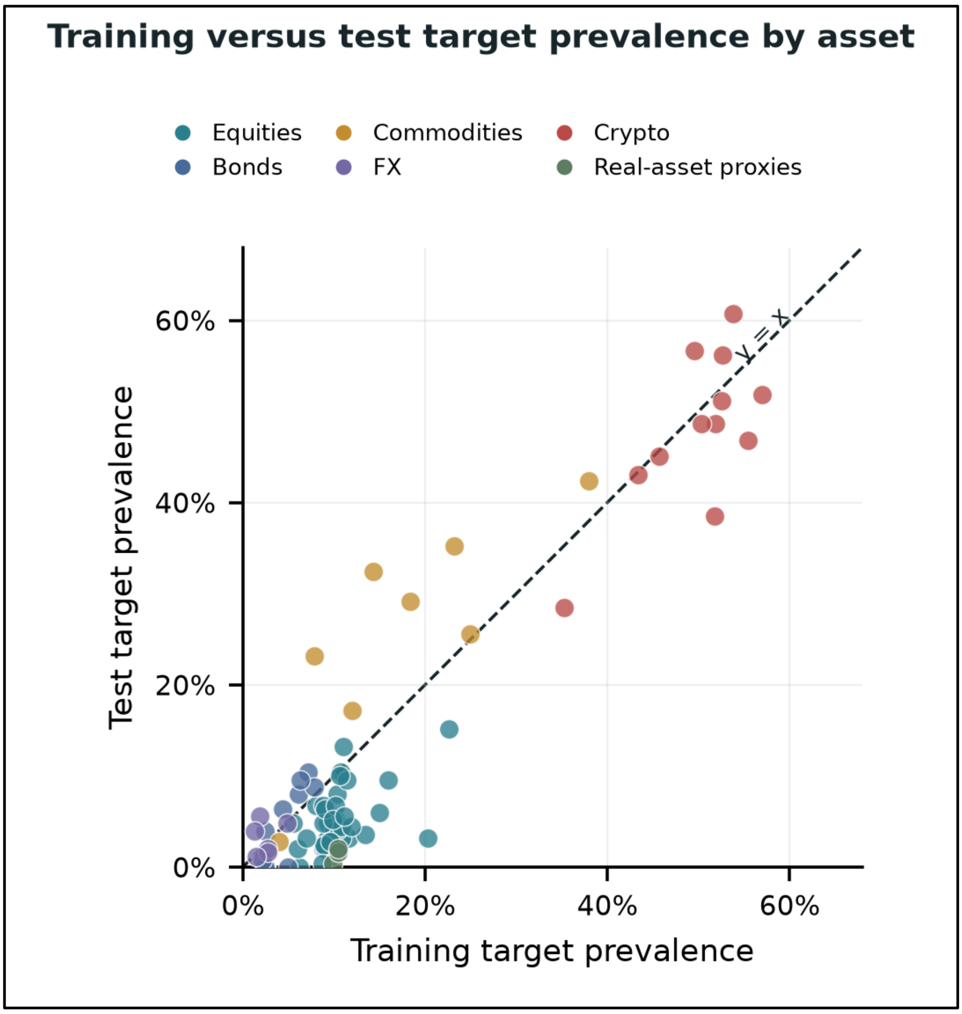

Figure A4. *Training versus test target prevalence by asset for the 79-asset model-evaluation endpoint population. The 45° reference line marks equal prevalence across the two periods; points are coloured by asset family.*

These prevalence plots complement the static-prior comparison in the main text. A stable ordering of asset or family event rates can support strong pooled discrimination even when a model contributes little within-asset timing information. The plots should therefore be interpreted alongside within-asset ROC-AUC, identity interventions and chronological-order sensitivity rather than as standalone evidence.

Appendix B. Controlled simulation and recovery evidence

The controlled simulation tests the diagnostic logic under mechanisms known by construction. World A contains heterogeneous asset priors without an ordered temporal signal; World B removes prior heterogeneity and plants ordered temporal information. The recovery tests are bounded secondary checks on the opened historical period.

Table B1. Controlled simulation mechanism estimates

Mechanism	Estimate	95% interval	Interpretation
Static-prior pooled AUC increase	0.400021	[0.391244, 0.408797]	Static heterogeneity can inflate pooled discrimination.
No-signal within-asset AUC	0.500086	[0.497032, 0.503141]	Within-asset ranking remains at chance without ordered signal.
Strong-signal within-asset AUC	0.783997	[0.781142, 0.786852]	The deliberately planted ordered signal is recovered.
Strong-signal reversal loss	0.380991	—	Reversing chronology is costly when order matters.
Strong-signal permutation loss	0.102297	—	Permutation also degrades the planted temporal signal.

Table B2. Bounded recovery tests

Recovery design	Primary result	Decision
Static + dynamic residual	Within-asset AUC 0.525360; interval [0.443706, 0.602033]	Paired-improvement and chronology checks failed.
Within-asset objective Transformer	Within-asset AUC 0.561924; interval [0.492527, 0.625022]	Uncertainty, paired-improvement and chronology checks failed.
Continuous downside Transformer	Equal-asset MAE 0.037816	Asset mean 0.020896 and ridge 0.020691 were stronger.

These supplementary results do not create new headline findings. They show that the main measurement conclusion survived three targeted redesigns, while the controlled simulation demonstrates that the diagnostic suite can recover ordered temporal structure when it is deliberately present. The simulation validates the behaviour of the diagnostics under known conditions; it is not external market replication or an estimate of real-market effect sizes.

Appendix C. Reproducibility specification

This appendix consolidates the fixed implementation details most likely to matter for independent reconstruction. It supplements, rather than replaces, the methodological description in the main text.

Table C1. Evaluation and score conventions

Item	Fixed specification
Forecast window	60 observed sessions ending at t
Target horizon	Observed sessions t+1 to t+10
Train	2010-01-04 to 2023-11-24; 245,055 windows
Validation	2023-12-14 to 2025-02-23; 20,494 windows
Test	2025-03-15 to 2026-06-13; 21,514 windows
Boundary controls	Purge 18 pooled-calendar dates; embargo 1 date; zero audited crossings
Numerical inputs	34 = 27 engineered market + 7 macro/context
Asset input	12-D learned asset embedding repeated across timesteps
Conditioned channels	46 after concatenation
Ranking scores	Raw ensemble scores for ROC-AUC, PR-AUC and within-asset AUC
Probability/threshold scores	Validation-selected isotonic probabilities for Brier, log loss, F1 and balanced accuracy
Neural seeds	7, 42, 123

The public repository provides the exact configured instrument universe, provider symbols, acquisition code, frozen compact evidence, tests and dissertation build materials. Raw provider data, model checkpoints and private prediction artefacts are not redistributed. The scientific claims therefore rely on the frozen evidence chain and documented reconstruction routes rather than redistribution of provider-restricted data.

Appendix D. Feature and data terminology

This appendix defines finance-specific indicators and macroeconomic series used in the feature set. Definitions are intentionally concise and describe how each term is interpreted in this study rather than providing a general finance textbook treatment.

Term / series	Meaning in this study
Adjusted close	Provider-adjusted closing price that accounts for corporate actions such as splits and distributions. It is used for returns and target construction.
Simple return	Percentage price change over the stated interval.
Log return	Natural logarithm of the price ratio between consecutive observations.
Overnight gap	Price change from the previous close to the next observed open.
Intraday range proxy	A same-session measure derived from the observed high and low to represent within-session price range.
Rolling volatility	Variability of recent returns calculated over a fixed trailing window.
Downside volatility	Volatility measure that emphasises negative-return observations.
EWMA volatility	Exponentially weighted volatility estimate that places greater weight on more recent returns.
Cumulative return	Total return accumulated over the stated trailing window.
Moving-average distance	Difference between price and a trailing moving-average reference, used as a trend measure.
5/20 crossover	Difference or relative position between short (5-session) and longer (20-session) moving averages.
RSI	Relative Strength Index; a bounded momentum indicator comparing the magnitude of recent gains and losses.
MACD	Moving Average Convergence Divergence; a momentum/trend indicator based on the difference between faster and slower exponential moving averages.
Volume z-score	Current volume standardised relative to its recent rolling mean and standard deviation.
Volume/average ratio	Current volume divided by a recent moving-average volume benchmark.
Drawdown	Decline from a previous rolling high to the current price.
Maximum adverse excursion	Largest fall relative to the forecast-date price during the future target horizon. The study uses the minimum origin-relative return over t+1 to t+10.
Future realised volatility	For the target, the root-sum-square of returns over t+1 to t+10.
Trailing volatility benchmark	Rolling sample standard deviation of the previous 20 returns (ddof=1), scaled by $\sqrt{10}$.
Target prevalence	Fraction of eligible observations whose binary adverse-event label equals 1 within an asset, family or split.
DFF	FRED effective federal funds rate series.
DGS2	FRED 2-year U.S. Treasury constant-maturity yield.
DGS10	FRED 10-year U.S. Treasury constant-maturity yield.
T10Y2Y	FRED 10-year minus 2-year U.S. Treasury yield spread.
VIXCLS	FRED series for the CBOE Volatility Index (VIX), a market-implied equity-volatility measure.
BAMLH0A0HYM2	FRED ICE BofA U.S. High Yield Index option-adjusted spread, used as a credit-risk context series.
DTWEXBGS	FRED nominal broad U.S. dollar index.

All seven FRED context series were lagged by at least one business day before alignment with market observations. The downloaded macroeconomic series were current-vintage data rather than a fully reconstructed point-in-time historical release database; this limitation is reported in the main discussion.

Appendix E. Secondary crypto-hourly volatility experiment

A separate exploratory track examined hourly cryptocurrency volatility forecasting. It is reported here to preserve the broader experimental record, but it is not part of the final daily multi-asset classification study and does not support the dissertation's main empirical claim.

Table E1. Scope and results of the separate hourly crypto track

Item	Secondary-track result
Universe	20 cryptocurrency pairs
Hourly observations	1,213,437 rows
Forecast task	4-hour log-realised-volatility forecasting
Model	Heteroscedastic bidirectional LSTM (BiLSTM)
Mean RMSE	0.6015 ± 0.0049
Pearson correlation	0.5616
Spearman correlation	0.5289
Uncertainty output	Exploratory prediction intervals behaved sensibly, but were not part of the final falsification framework

The hourly experiment used a different sampling frequency, prediction target and model objective from the final daily adverse-event classifier. It did not undergo the same corrected split audit, identity interventions, within-asset evaluation, chronology perturbations or controlled simulation battery. It should therefore be treated as a separate exploratory result rather than a positive exception to the main dissertation conclusion.

Earlier project-wide references to approximately 1.57 million observations aggregated distinct daily and hourly experimental tracks. That figure should not be interpreted as the size of one unified dataset or one model-training sample. The headline dissertation evidence is based on the final daily multi-asset study described in the main text.

No additional headline result is promoted from this appendix. Its purpose is transparency about the broader experimental programme and preservation of a technically distinct volatility-forecasting track.

Appendix F. Generative AI Statement

Generative AI tools were used as permitted support during the project for drafting/language editing, document formatting, code-structure review and troubleshooting. The research direction, dataset selection, experimental interpretation, final claims and submitted work were reviewed and approved by the student. The student retained responsibility for the research question, dataset selection, experimental design, execution of the analysis, interpretation of the results, and all final claims made in the dissertation. AI-assisted outputs were critically reviewed before use. Where relevant, numerical claims and generated summaries were checked against the project's reproducible code, automated tests, frozen result tables and CSV artefacts before inclusion in the dissertation. Generative AI outputs were not treated as scientific evidence in their own right.